

Programming in Python for Friends and Relatives: Part 9

Scripts

for Home Network

Some nifty scripts to check remote systems over the Internet.

Thanks to broadband, most homes have a computer network. However, I often get a call from my parents on how their computer is not working. This, in their minds, is the root cause for a vast variety of issues they face. Wouldn't it be nice if I could just sign into their computer and examine the problem? Do you know that with the availability of 'broadband', this is, indeed, viable?

Python has excellent modules for network programming—the key modules are `socket` and `threading`. However, to monitor a network, you will need to write small scripts for specific tasks and not worry about the intricacies of network programming. So, you can feel confident about encouraging your relatives to switch to Linux *and* offer support to them remotely!

To log in to a remote computer, you will need to know its IP address. On a broadband connection, the IP address is typically not fixed. Once your script finds the IP, you have to be informed of it. The easiest way is for the script to mail it to you.



Note: The code I'll present has been tested on Fedora 9, but should work unchanged on Ubuntu and most other distributions as well.

Finding the IP address of the machine

Let us suppose the broadband modem is connected using the `pppoe` protocol, which is easier to handle. In this case the public IP is on the local system. You can find that out by using the `/sbin/ifconfig` command. You should see an entry like `ppp0` in addition to entries for `lo` and `eth0`. You are interested in the `ppp0` entry and, in particular, the line

related to the 'inet addr' entry. So, here is our little script using the versatile subprocess module added in Python 2.4:

```
from subprocess import Popen, PIPE
def get_ip():
    proc = Popen(['/sbin/ifconfig'], stdout=PIPE, stderr=PIPE)
    output = proc.communicate()
    if output[1] != "":
        response = 'Error: ' + output[1]
    else:
        response = [line for line in output[0].split("\n") if 'inet addr:' in line]
    return str(response)
print 'The IP is: ' + get_ip()
```

The above script opens a pair of pipes to the `ifconfig` command and passes 'ppp0' as a parameter. The `communicate` method returns a tuple for the `stdout` and `stderr` of the command. In this

instance, if you find that `stderr` is not empty there must have been an error; otherwise, you split the `stdout` on a new line and select the line containing the IP address.

A second possibility is that the modem is set up as a router. So, the external IP is on the router. The home models come with a Web interface to the modem. You can call the relevant page using

